Suraj Krishna Banavalikar

+91 9663132144 skbanavalikar2105@gmail.com surajkb2005 github.com/surajkb2005 Portfolio

Technical Skills

Languages: Java, C/C++, Python, JavaScript
Frontend: HTML, CSS, React.js
Backend: Node.js, Express.js, REST APIs
Databases: MongoDB, PostgreSQL, MySQL, Pinecone
Developer Tools: Git, GitHub, Docker, Azure & AWS cloud, Linux, Prometheus, Grafana, Postman, Power BI

Education

Ramaiah Institute of Technology, Bangalore **Oct 2023 – Present**
B.E. in Computer Science and Engineering *CGPA: 9.19/10*

St. Joseph's PU College, Karwar **Aug 2021 – Mar 2023**
PCMC (Physics, Chemistry, Mathematics, Computer Science) *Grade: 96.16%*

Relevant Coursework

- Data Structures & Algorithms
- Operating System
- Object Oriented Programming
- Computer Networks
- Database Management Systems
- Software Engineering
- Artificial Intelligence
- DevOps and Cloud
- Distributed Systems

Projects

Campus Helpdesk | *Node.js/Express, Python, REST APIs, MongoDB, Pinecone, JavaScript, Docker* **Showcase**

- Developed an AI–powered Campus Helpdesk using Retrieval–Augmented Generation (RAG), vector search, and hybrid re-ranking with Pinecone.
- Implemented JWT authentication, role–based authorization, and RESTful APIs for secure access to campus resources.
- Secured the backend with configurable CORS, API rate limiting and scaled with PM2 clustering to support ∼150 concurrent users.
- Optimized performance with MongoDB connection pooling and asynchronous job queues (20 jobs per batch with exponential backoff), reducing API blocking and improving throughput under concurrent load.
- Load–tested at 50–60 concurrent users with stable latency, and reduced manual query–handling effort by ∼40% based on evaluation against real usage.
- Containerized micro services with Docker, enabling consistent deployments.

Zero–Server AI Privacy Gateway | *JavaScript, IndexedDB* **GitHub**

- Built a Manifest V3 Chrome extension that intercepts outgoing ChatGPT prompts by patching `fetch`, redacts PII locally, and sends only placeholders, so no raw data reaches the LLM provider and no backend is needed.
- Built a PII detection pipeline using regex, Luhn validation and a confidence-scored heuristic for 22 data types; compared it with a C++/WebAssembly detector.
- Stored placeholder mappings in IndexedDB and swapped the original values back into responses using a MutationObserver.
- Evaluated on a 2,400-prompt synthetic benchmark (52 categories, including adversarial look-alikes): 93.7% recall, 74.7% precision, F1 0.83, with ∼16 ms average latency per prompt.

StellarRoute | *Python, Leaflet.js, REST APIs, Redis, Docker, Prometheus, Grafana* **GitHub | Live**

- Built a GPS–denied navigation fallback using smartphone IMU sensors streamed over WebSockets, with dead–reckoning position updates on a live map during simulated GPS outages caused by solar storms.
- Integrated NOAA SWPC live Kp index and solar–wind data to generate GPS–error risk heatmaps and storm–aware A* routing; designed a provider–based architecture to add ISRO Aditya–L1 archival data (PRADAN) as a future source.
- Containerized the app with Docker and set up a GitHub Actions CI/CD pipeline that deploys behind an Nginx reverse proxy.

Experience

RIT S.T.A.R.D.U.S.T Club **May 2025 – Present**
Software Engineer — Avionics Member

- Participated in the IRoC–U 2026 competition by developing autonomous drone simulations using ROS 2 and Gazebo.
- Assisted in validating simulated drone behavior for competition preparation.

Awards and Certifications

wHACKiest 2025 — 2nd Place

- Secured **2nd Place** among **100+ teams** at the 24–hour intra–college hackathon by developing **StellarRoute** as the Backend Engineer. **Certificate**